01 · REGISTRY — nothing can be named that was not first bound to a code

a variable declaration

```
name acr_mg_g
type real | int | bool | code | date
unit mg/g
binding system | code | version
role observation | action
codes the value-set expansion
plausible a physiological range
unbound true = a named hole
```

role is why an obligation must constrain something a care plan can actually change.

as the binder emits it

```
VarDef(name="acr_mg_g", type="real",
 unit="mg/g",
 binding="http://loinc.org|9318-7|2.78",
 role="observation",
 plausible=(0, 3000))
```

LOINC 14585-4 is the same ratio in mg/mmol. The two score identically on every text signal and are a factor of 8.84 apart, so the unit in the declaration is part of the meaning.

events, and the two kinds of hole

```
EventDef("potassium_check",
 binding="http://loinc.org|2823-3|2.78")
```

```
VarDef("has_high_risk", unbound=True)
EventDef("start_lrd", unbound=True)
```

a predicate, and an anchor, the sentence names

A hole is a real free variable: no binding, no data source, and no domain axiom — constraining it would assert something the guideline never said. It is a property of a declaration, not a new node.

`the decoder's variable and event enums ARE these lists of names — an unbound concept is not expressible, only refusable`

02 · GRAMMAR — four leaves, three connectives — and that is all of it

LEAVES

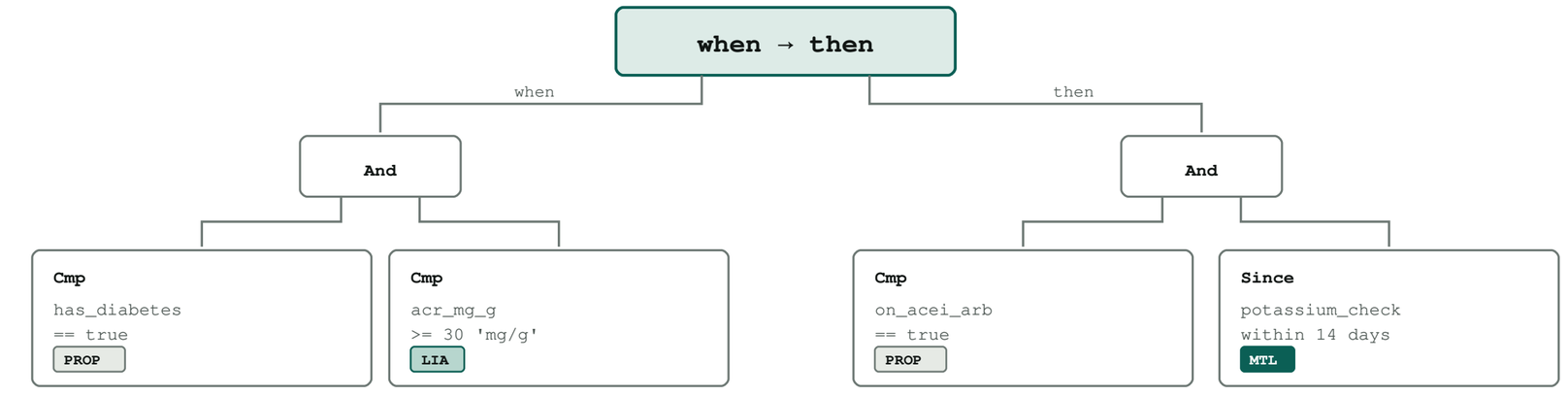
```
Cmp(var, op, value)                          PROP  LIA
Between(var, lo, hi, lo_incl, hi_incl)       LIA
Since(event, days)                           MTL
Delta(var, baseline, direction,              LIA
 lo_pct, hi_pct)
```

CONNECTIVES

```
Not(e) And([e, …]) Or([e, …])
```

Or is mandatory, not a convenience. An expression language without disjunction does not remove disjunctions — it converts them into conjunctions. One AND-for-OR substitution gave the wrong referral verdict across 69.5% of the plausible patient box, and it typechecked.

"In patients with diabetes and an ACR ≥ 30 mg/g, treat with an ACEi or ARB and check serum potassium within 14 days."



`fragment_of walks the tree and returns the weakest fragment that contains it — here the maximum is MTL.`
Computed from the tree, never inferred from variable names, so the formalizability spectrum is a function of the output rather than of an annotator.

03 · THE RULE — a formula under an operator, not a labelled action

the whole of it

φ = (when → then)

```
under a deontic operator:
obligation O(then | when)
permission P(then | when)
```

Prohibition is not a third case: F(p) = O(¬p), a negated consequent.

carried alongside, and carrying no semantics

```
action recommend | avoid | hold |
 monitor | refer | screen
target "an ACEi or ARB"
strength "1B"
source page + the verbatim sentence
```

These exist so a reviewer can check the formula against the sentence. Branching on action in verification code means something has gone wrong.

refusal is a value of the same decode

```
outcome ∈ { "rule", "refusal" }
```

```
reason_code ∈ { vague_predicate, defeasible,
 not_patient_level, unsupported_temporal,
 unbound_concept, beyond_dsl }
```

A discriminator rather than a top-level union, because the decoder cannot express oneOf. The model still has to commit to one or the other — abstaining is an answer, not a failed call.